

Disproof of Conjecture 00000001141: the centre of $u(\mathfrak{sl}_n)$ has Krull dimension 0, not $n - 1$

jilint777

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Abstract

Conjecture 00000001141 asserts that the Krull dimension of the centre of the restricted enveloping algebra $u(\mathfrak{sl}_n)$ is $n - 1$, and that for every reductive $\mathfrak{g}$ the centre of $u(\mathfrak{g})$ is spanned by the Weyl-invariant polynomial ring of the p -centre, with dimension equal to the rank. The first clause is false for every $n \geq 2$ and every prime p . The algebra $u(\mathfrak{g})$ has finite dimension $p^{\dim \mathfrak{g}}$, so its centre is a finite-dimensional commutative algebra. Every prime ideal of such an algebra is maximal, so its Krull dimension is $0 \neq n - 1$. We make this concrete for $n = 2$, $p = 3$. The algebra $u(\mathfrak{sl}_2)$ has dimension 27. Its centre Z has dimension 4 and 81 elements, with $Z \cong \mathbb{F}_3 \times \mathbb{F}_3[x, y]/(x, y)^2$. It has exactly two prime ideals, both maximal. The Lean 4 development (core library only) builds a model of $u(\mathfrak{sl}_2)$ over $\mathbb{F}_3$ from scratch. It proves the defining relations and that the model has exactly 3^{27} elements, and it proves from first principles that its centre has Krull dimension 0; in particular this dimension is not $2 - 1$. It also shows that the centre is not one-dimensional, so the “dimension equals the rank” part of the second clause fails for $\mathfrak{g} = \mathfrak{sl}_2$ as well. A standard-library Python script recomputes everything for $p = 2, 3, 5$ by a different method. We also discuss the other readings. For the full enveloping algebra $U(\mathfrak{sl}_n)$ in characteristic p , the Krull dimension of the centre is $n^2 - 1$, which again is not $n - 1$. Only the characteristic-0 statement for $U(\mathfrak{sl}_n)$ gives $n - 1$ (Harish-Chandra), and that statement is true but is not the one made.

1 The statement

Definition: The restricted enveloping algebra $u(\mathfrak{g})$ of a restricted Lie algebra $\mathfrak{g}$. *Conjecture:* The Krull dimension of the center of $u(\mathfrak{sl}_n)$ is $n - 1$ (generated by the p -centers of the various $\mathfrak{sl}_2$ -sections), and for every reductive $\mathfrak{g}$ the center of $u(\mathfrak{g})$ is spanned by the Weyl-invariant polynomial ring of the p -center, with dimension always equal to the rank.

The Chinese version says the same thing. The conjecture is a conjunction, so it is enough to refute the clause

$$(C1) \quad \text{Kdim } Z(u(\mathfrak{sl}_n)) = n - 1 \quad (\text{for every } n),$$

and we refute it at $n = 2$: the claim is $\text{Kdim } Z(u(\mathfrak{sl}_2)) = 1$. Section 7 shows that the second clause fails for $\mathfrak{g} = \mathfrak{sl}_2$ as well.

2 Background

Let k be a field of characteristic $p > 0$. A *restricted Lie algebra* is a Lie algebra $\mathfrak{g}$ over k with a p -map $x \mapsto x^{[p]}$ such that $\text{ad}(x^{[p]}) = (\text{ad } x)^p$ (plus the usual axioms). The *restricted enveloping*

algebra is

$$u(\mathfrak{g}) = U(\mathfrak{g}) / (x^p - x^{[p]} : x \in \mathfrak{g}).$$

It suffices to take x in a basis of $\mathfrak{g}$. Jacobson's PBW theorem for $u(\mathfrak{g})$ states that if $x_1, \dots, x_d$ is a basis of $\mathfrak{g}$, the ordered monomials $x_1^{a_1} \cdots x_d^{a_d}$ with $0 \leq a_i < p$ form a basis of $u(\mathfrak{g})$. In particular $\dim_k u(\mathfrak{g}) = p^{\dim \mathfrak{g}}$ (see Jacobson, *Lie Algebras*, Ch. V, or Strade–Farnsteiner, *Modular Lie Algebras and their Representations*, Ch. 2).

For $\mathfrak{sl}_n$ the p -map is the matrix p -th power. On the basis $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ of $\mathfrak{sl}_2$ it is $e^{[p]} = 0$, $f^{[p]} = 0$, $h^{[p]} = h$. For p odd the centre of $\mathfrak{sl}_2$ is 0, so this p -map is the only one. Hence

$$u(\mathfrak{sl}_2) = k\langle e, h, f \rangle / ([h, e] = 2e, [h, f] = -2f, [e, f] = h, e^p = 0, f^p = 0, h^p = h).$$

The *Krull dimension* of a commutative ring R is the supremum of the lengths k of chains $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_k$ of prime ideals.

3 The first clause is false for all $n \geq 2$

Lemma 1. *Let Z be a nonzero commutative algebra of finite dimension over a field k . Every prime ideal of Z is maximal, and $\text{Kdim } Z = 0$.*

Proof. Let P be prime. Then $D = Z/P$ is a finite-dimensional k -algebra without zero divisors. For $0 \neq x \in D$, multiplication by x is an injective k -linear endomorphism of D , hence surjective, so $xy = 1$ for some y . So D is a field and P is maximal. Two maximal ideals are never strictly contained in one another, so there is no chain of primes of length ≥ 1 . A maximal ideal exists because $Z \neq 0$, so $\text{Kdim } Z = 0$. $\square$

Theorem 2. *For every field k of characteristic $p > 0$ and every $n \geq 1$, $\text{Kdim } Z(u(\mathfrak{sl}_n)) = 0$. In particular $\text{Kdim } Z(u(\mathfrak{sl}_n)) \neq n - 1$ for all $n \geq 2$.*

Proof. $u(\mathfrak{sl}_n)$ has dimension p^{n^2-1} , so its centre is a finite-dimensional commutative algebra. It contains $k \cdot 1 \neq 0$. Apply Lemma 1. $\square$

The same argument applies to every finite-dimensional algebra. This covers $u(\mathfrak{g})$ for any restricted $\mathfrak{g}$ and the reduced enveloping algebras $u_\chi(\mathfrak{g})$. When $k = \mathbb{F}_q$ is finite, the centre is a finite ring. For finite rings the proof below is the one formalized in Lean. It uses only that powers of an element repeat.

Lemma 3 (finite rings). *Let R be a finite commutative ring and $P \subseteq Q$ prime ideals. Then $P = Q$.*

Proof. Let $x \in Q$. By the pigeonhole principle $x^i = x^j$ for some $i < j$. Put $d = j - i$; then $x^{m+d} = x^m$ for all $m \geq i$, so $y = x^n$ with $n = d(i + 1)$ satisfies $n \geq 1$ and $y^2 = y$. Hence $y(1 - y) = 0 \in P$, so $y \in P$ or $1 - y \in P$. If $1 - y \in P \subseteq Q$, then since $y \in Q$ also $1 = y + (1 - y) \in Q$, which is impossible. So $x^n = y \in P$, and since P is prime, $x \in P$. $\square$

4 The example $n = 2, p = 3$ in detail

4.1 Normal ordering

Lemma 4. *In any associative algebra generated by e, h, f with $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$, for all $a \geq 0$ and every polynomial q :*

$$h e^a = e^a(h + 2a), \quad q(h) e = e q(h + 2), \quad f q(h) = q(h + 2) f, \quad [e^a, f] = a e^{a-1}(h + a - 1).$$

Consequently, for the monomials $m_{abc} = e^a h^b f^c$,

$$\begin{aligned} e \cdot m_{abc} &= m_{a+1,b,c}, & h \cdot m_{abc} &= e^a(h + 2a)h^b f^c, \\ f \cdot m_{abc} &= e^a(h + 2)^b f^{c+1} - a e^{a-1}(h + a - 1)h^b f^c. \end{aligned}$$

Proof. $he = e(h + 2)$ is $[h, e] = 2e$, and induction on a gives the first formula. The second follows from it, and the third is the mirror image $fh = (h + 2)f$ of $[h, f] = -2f$. For the last, $[e^{a+1}, f] = e[e^a, f] + [e, f]e^a = a e^a(h + a - 1) + h e^a = e^a(a(h + a - 1) + h + 2a) = (a + 1)e^a(h + a)$. The displayed products follow. $\square$

Proposition 5 (PBW spanning). *If B is an associative algebra generated by e, h, f satisfying the relations of $u(\mathfrak{sl}_2)$, then B is spanned by the p^3 monomials m_{abc} , $0 \leq a, b, c < p$. Hence $\dim u(\mathfrak{sl}_2) \leq p^3$.*

Proof. Let S be the span of the m_{abc} . It contains 1. By Lemma 4, together with $e^p = f^p = 0$ and $h^p = h$ (which reduce every polynomial in h to degree $< p$), S is stable under left multiplication by e, h and f . So S is a left ideal containing 1, and $S = B$. $\square$

(The reverse inequality is Jacobson's theorem. It is also reproved below for $p = 3$, because our model has exactly 3^{27} elements.)

4.2 A concrete model of $u(\mathfrak{sl}_2)$ over $\mathbb{F}_3$

Let $V = \mathbb{F}_3^{27}$ with basis b_i , $i = 9a + 3b + c$, standing for m_{abc} . Let $E, H, F \in \text{End}(V)$ be the operators given by the formulas of Lemma 4, i.e. left multiplication by e, h, f on $u(\mathfrak{sl}_2)$. Let R_E, R_H, R_F be the right multiplications. All six are explicit sparse 27×27 matrices over $\mathbb{F}_3$. Let $A \subseteq \text{End}(V)$ be the unital subalgebra generated by E, H, F .

Proposition 6. 1. E, H, F satisfy $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$, $E^3 = 0$, $F^3 = 0$, $H^3 = H$.

2. The evaluation map $\text{ev}: A \rightarrow V$, $\varphi \mapsto \varphi(b_0)$, is bijective. So $|A| = 3^{27}$.

3. Consequently $A \cong u(\mathfrak{sl}_2)$ over $\mathbb{F}_3$.

Proof. (1) All maps involved are linear, so it suffices to compare them on the 27 basis vectors. This is a finite computation (done by `decide` in Lean).

(2) Surjectivity: $E^a H^b F^c(b_0) = b_{9a+3b+c}$ for all $a, b, c < 3$ (finite computation), and A is closed under linear combinations. Injectivity: R_E, R_H, R_F commute with E, H, F (finite computation), hence with all of A . Also $b_{9a+3b+c} = R_F^c R_H^b R_E^a(b_0)$ (finite computation). So for $\varphi \in A$ we get $\varphi(b_i) = R_F^c R_H^b R_E^a(\varphi(b_0))$, and φ is determined by $\varphi(b_0)$.

(3) By (1) and the universal property of $U(\mathfrak{sl}_2)$ and of its quotient $u(\mathfrak{sl}_2)$, there is a surjective algebra map $u(\mathfrak{sl}_2) \rightarrow A$ sending $e, h, f \mapsto E, H, F$. By Proposition 5, $|u(\mathfrak{sl}_2)| \leq 3^{27} = |A|$, so the map is bijective. $\square$

4.3 The centre

Gaussian elimination over $\mathbb{F}_3$ (in `verify.py`) shows that $Z = Z(u(\mathfrak{sl}_2))$ has dimension 4, with basis

$$1, \quad z_2 = h + h^2 + ef, \quad z_3 = 2h + 2h^2 + 2ehf + e^2f^2, \quad z_4 = 2e^2hf^2 + e^2h^2f^2.$$

The Casimir element is $C = ef + fe + \frac{1}{2}h^2 = ef + fe + 2h^2 = 2z_2$. The products are $z_2^2 = z_3$, $z_2z_3 = -z_3$, $z_3^2 = z_3$ and $z_2z_4 = z_3z_4 = z_4^2 = 0$. So $\varepsilon = z_3$ is an idempotent, $Z\varepsilon = \mathbb{F}_3\varepsilon$, and $Z(1 - \varepsilon)$ is spanned by $1 - \varepsilon$, $x = z_2 + z_3$ and $y = z_4$, with $x^2 = xy = y^2 = 0$. Therefore

$$Z \cong \mathbb{F}_3 \times \mathbb{F}_3[x, y]/(x, y)^2.$$

This ring has exactly two prime ideals, $\mathbb{F}_3\varepsilon \oplus (x, y)$ and $Z(1 - \varepsilon)$. Each has 27 elements and both are maximal. The first is the kernel of the trivial character $e, h, f \mapsto 0$. There is no chain $P \subsetneq Q$, so $\text{Kdim } Z = 0 \neq 1$. A brute-force search in `verify.py` over all 212 subspaces of Z confirms that Z has 14 ideals and 2 primes, with no inclusions between the primes. For comparison, the script also computes $\dim Z(u(\mathfrak{sl}_2)) = 5, 4, 7$ for $p = 2, 3, 5$, with 2, 2, 3 prime ideals respectively.

5 The Lean formalization

The directory `lean4/` contains a Lean 4.19.0 project that uses only the core library. All objects are defined from scratch.

- **F3** (the field $\mathbb{F}_3$), **V** = $\mathbb{F}_3^{27}$ (functions $\text{Fin } 27 \rightarrow \text{F3}$), basis vectors **bv**, finite sums, and additive maps. **ext_basis**: two additive maps that agree on the basis are equal.
- **opE**, **opH**, **opF**, **opRE**, **opRH**, **opRF**: the six sparse operators (data tables). **Gen**: the inductive closure of id, E, H, F under $+$, $-$ and composition, i.e. the algebra A .
- **relations**: Proposition 6(1). **ev_bijective**: Proposition 6(2). Part (3), via Proposition 5 and the universal property, is argued in this report and not in Lean.
- **CRing**: commutative rings, with **IsIdeal**, **IsPrime**, **HasChain k** (a chain of primes $P_0 \subsetneq \dots \subsetneq P_k$), and **KrullDimIs d** (**HasChain d** holds and every chain has length $\leq d$, i.e. the supremum is d). **Finite** means there is an injection into a type in which the image is covered by a list.
- **primes_incomparable** (Lemma 3), **finite_krullDim_zero**, **finite_krullDim_ne**, and **clause_fails_for_finite_rings**: no finite commutative ring has Krull dimension $n - 1$ for $n \geq 2$.
- **ZU**: the centre $\{\varphi \in A : \varphi\psi = \psi\varphi \ \forall \psi \in A\}$, with the commutative ring structure **ZRing**. **ZRing_finite**: it is finite, by injectivity of **ev**. **P0_prime**: the kernel of the trivial character $\chi(\varphi) = \text{coefficient of } b_0 \text{ in } \varphi(b_0)$ is a prime ideal.
- Main results: **krullDim_center** : **ZRing.KrullDimIs 0**, **conjecture_00000001141_false** : $\neg \text{Clause } 2$, where **Clause n** := **ZRing.KrullDimIs (n - 1)**, and **krullDim_center_ne**: no $d \geq 1$ is the Krull dimension.
- **center_not_rank_dim**: the Casimir operator $EF + FE + 2H^2$ is central (checked on generators, then by induction), and $Z(A)$ is not of the form $\mathbb{F}_3 \cdot z$, so $\dim_{\mathbb{F}_3} Z \geq 2 > 1 = \text{rank } \mathfrak{sl}_2$.

The build takes under a minute. The `#print axioms` report shows only `propext`, `Quot.sound` and (for the pigeonhole step, which uses `by_cases`) `Classical.choice`. There is no `sorry` and no `native_decide`. The operator tables are data. Their meaning is fixed by the theorems `relations` and `ev_bijective`, and `verify.py` re-derives them independently by word rewriting.

Non-vacuity. `HasChain 0` holds through the explicit prime `P0`, so the Krull dimension is exactly 0 rather than undefined. The centre is a genuine subring of A , with $\dim_{\mathbb{F}_3} Z \geq 2$ shown in Lean and $= 4$ by `verify.py`. The relations show that A is a model of $u(\mathfrak{sl}_2)$, not of some smaller quotient.

6 Readings of the statement

1. **Literal reading** ($u(\mathfrak{sl}_n)$, any field of characteristic p , any p). Here $\text{Kdim } Z = 0$ by Theorem 2, while the claim is $n - 1$. The claim fails for every $n \geq 2$; it holds trivially for $n = 1$, where $\mathfrak{sl}_1 = 0$. The Lean proof treats $n = 2$, $p = 3$, $k = \mathbb{F}_3$. Over an extension $K \supseteq \mathbb{F}_3$ the centre is $Z \otimes_{\mathbb{F}_3} K$, since the centre is the kernel of a linear map, and it is still finite-dimensional.
2. **Vector-space dimension instead of Krull dimension.** $\dim Z(u(\mathfrak{sl}_2)) = 5, 4, 7$ for $p = 2, 3, 5$, never 1. Lean proves $\dim \geq 2$ for $p = 3$. So the clause fails under this reading too.
3. **$U(\mathfrak{sl}_n)$ in characteristic p** (in case “ u ” was meant as U). The p -centre $Z_p = k[x^p - x^{[p]} : x \in \text{basis}]$ is a polynomial ring in $n^2 - 1$ variables, and $U(\mathfrak{sl}_n)$ is a free Z_p -module of rank p^{n^2-1} (Zassenhaus 1954; see Jantzen, *Representations of Lie algebras in prime characteristic*, NATO ASI Ser. C 514, 1998). So $Z_p \subseteq Z(U)$ and $Z(U)$ is a finitely generated Z_p -module (a submodule of a finite module over a Noetherian ring). Hence $Z(U)$ is integral over Z_p , and $\text{Kdim } Z(U) = \text{Kdim } Z_p = n^2 - 1 \neq n - 1$ for $n \geq 2$. For $n = 2$ the value is 3. `verify.py` checks that $e^p, f^p, h^p - h$ are central in $U(\mathfrak{sl}_2)$ for $p = 2, 3, 5$. This reading is in the report only.
4. **Characteristic 0, $U(\mathfrak{sl}_n)$.** By Harish-Chandra and Chevalley, $Z(U(\mathfrak{sl}_n)) \cong S(\mathfrak{h})^W$ is a polynomial ring in $n - 1$ variables, so its Krull dimension is $n - 1$. This true statement is probably what the conjecture garbles. But the conjecture explicitly concerns the *restricted* enveloping algebra and the p -centre, which exist only in characteristic p . The statement must be judged as written, and as written it is false.

7 The second clause

For reductive $\mathfrak{g}$ the clause asserts that $Z(u(\mathfrak{g}))$ is spanned by “the Weyl-invariant polynomial ring of the p -centre”, with dimension equal to the rank. In $u(\mathfrak{g})$ the p -centre generators $x^p - x^{[p]}$ are 0, so the p -centre maps onto $k \cdot 1$, and polynomials in it span only the scalars. For $\mathfrak{g} = \mathfrak{sl}_2$ (reductive, rank 1) the centre contains the non-scalar Casimir element. Its dimension is $4 \neq 1$ for $p = 3$ (≥ 2 in Lean), and its Krull dimension is $0 \neq 1$. So the second clause is false as well, under either meaning of “dimension”. Refuting one clause is already enough.

8 Independent verification

`verify.py` (standard library) rebuilds $u(\mathfrak{sl}_2)$ for $p = 2, 3, 5$ by rewriting words in e, h, f with the defining relations. It then checks:

- associativity (all triples for $p \leq 3$; for $p = 5$, $(gx)y = g(xy)$ for generators g);
- the relations, and that $m_{abc} = e^a h^b f^c$;
- the centre: dimension, closure and commutativity, and that the Casimir is central;
- the prime ideals: brute force for $p = 2, 3$, and nilradical plus Frobenius for $p = 5$;
- that the p -centre is central in $U(\mathfrak{sl}_2)$;
- that the six Lean tables equal the rewriting result.

All checks pass (see `verification.txt`).

9 Conclusion

$u(\mathfrak{sl}_n)$ is finite-dimensional, so its centre is Artinian and has Krull dimension 0. The claimed value $n - 1$ is wrong for every $n \geq 2$. For $n = 2$, $p = 3$ this is verified in Lean, down to an explicit model of $u(\mathfrak{sl}_2)$ and a from-scratch notion of Krull dimension. Conjecture 00000001141 is false.